

ARJUN VARMA

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EDUCATION

Columbia University Master of Science in Data Science

New York, NY
Expected Dec 2026

Teaching Assistant, Columbia Business School: Business Analytics II (Foundations of AI) and Hollywood and Big Data

Vellore Institute of Technology

Vellore, IN

Bachelor of Technology in Electronics and Communication Engineering

May 2022

Special Achiever Award | Merit Scholarship

WORK EXPERIENCE

Novo Nordisk

Plainsboro, NJ

Data Scientist Intern

Jun 2026 - Aug 2026

- Equipped field teams with explainable, tailored guidance by improving propensity scoring and next-best-engagement models and building an LLM layer for decision support across behavioral, access, competitive, and adoption signals.
- Identified and prioritized a targeted subset of U.S. HCPs to support pre-launch planning for an investigational hemophilia therapy using real-world claims data from Komodo Healthcare Map, business rules, and predictive signals.
- Deployed an XGBoost propensity model with 5.2x top-decile lift, turning profile, behavioral, access, and engagement signals into explainable priority tiers and a dashboard used by field teams and senior leadership nationwide.

ZS Associates

Pune, IN

Advanced Data Science Associate Consultant

Feb 2022 - Jun 2025

- Served as the direct technical point of contact for U.S.-based client directors and senior directors, translating ambiguous business needs into scoped analytics products, coordinating technical teams and external vendors, and guiding rollout and executive decisions.
- Cut weekly reporting from days to minutes for 100+ stakeholders across \$10B in oncology brands by standardizing IQVIA and Symphony Health claims, Flatiron Health EHR-derived data, and Texas Oncology data in a Spark/SQL data warehouse with product and territory KPIs.
- Recovered missing categorical labels in medical transaction data with positive-unlabeled classification, expanding oncology customer-journey coverage from ~40% to 95% across tumor types and territories.
- Productionized the models with MLflow tracking, feature/prediction drift monitoring, and CI tests, reducing silent failures.
- Saved ~50 hours per month and raised first-pass quality above 99% by leading a five-member legacy-rules modernization.
- Established audit-ready real-world evidence (RWE) analytics across \$4B+ in oncology products by engineering PySpark/SQL pipelines over millions of healthcare records with cohort and entity logic resilient to missing and miscoded fields.

PROJECT EXPERIENCE

Biliary Tract Cancer Early-Detection Model - Python, PySpark, scikit-learn, NLP

- Accelerated identification by ~45 days relative to standard diagnosis timing by combining clinical risk factors, K-means/GMM segmentation, and transformer-based NLP clustering.
- Scaled early-detection modeling across 250M longitudinal claims by building PySpark cohorts and clinical risk-factor features that supported patient segmentation and model scoring.
- Drove territory-level resource planning with adopted model outputs; methodology from the project was presented at the PMSA 2025 conference.

SunCulture: AI Product Classification for Credit Risk - Python, RAG, REST APIs

- Supported microloan creditworthiness assessment in East Africa by classifying 7M+ farmer transactions across 500+ product categories with a RAG-augmented system.
- Achieved 99% accuracy on a 10K-item holdout through hybrid rule-based and LLM-assisted classification, reducing manual review by 95%.

Filing Intelligence: Financial Research RAG - Python, FastAPI, ChromaDB, Next.js, GCP

[GitHub](#) / [Live Demo](#)

- Produced cited financial research across 15 companies and 4,967 passages from 128 filings, earnings decks, and call transcripts by deploying an evidence-first RAG workspace with scope parsing, Chroma retrieval, deterministic ranking, and bounded context.
- Operate reliable, auditable retrieval on Cloud Run with page/line citations and 72 automated checks; run authenticated FastAPI/Next.js services, a server-side identity-token boundary, cited-page PDF viewer, incremental sync, and freshness/readiness endpoints.

ClaimReady: AI Small-Claims Packet Generator - Python, OpenAI Agents SDK, FastAPI, Next.js, GCP

[GitHub](#) / [Live Demo](#)

- Generated a court-ready small-claims packet from contracts, invoices, emails, and images in under 60 seconds through a four-agent workflow with structured handoffs and multimodal extraction.
- Operationalized an auditable workflow on Cloud Run by integrating typed Pydantic outputs, legal RAG, live NY business-registry lookup, deterministic jurisdiction/damages tools, WebSocket traces, and ReportLab generation into FastAPI/Next.js services.

SKILLS

LLMs & Agents: RAG, LangChain, OpenAI Agents SDK, tool/function calling, structured outputs, multimodal workflows, prompt engineering

Retrieval & Evaluation: Embeddings, metadata filtering, evidence ranking, citations/provenance, deterministic and golden evals, guardrails

Programming: Python, SQL, R, TypeScript, JavaScript; pandas, NumPy, scikit-learn, PyTorch, XGBoost

Product Engineering: FastAPI, REST APIs, Server-Sent Events, WebSockets, Next.js, React, Chrome extensions

Data & MLOps: Spark/PySpark, Databricks, MLflow, production ML, ETL/ELT, data warehousing, Docker, CI/CD, GitHub Actions

Databases & Cloud: Postgres, Snowflake, DuckDB, ChromaDB; AWS; GCP Cloud Run, Cloud Storage, Firebase

Healthcare & RWE: RWD/RWE, longitudinal claims, EHR data, cohorts, patient journeys, HCP targeting, launch analytics